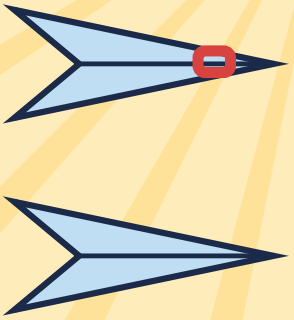
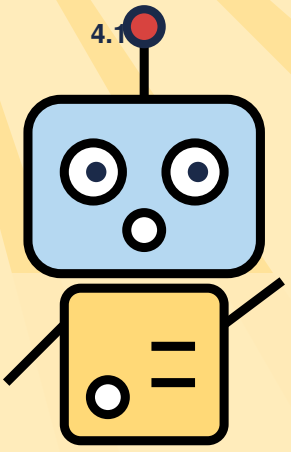
SCIENCE
Book 2

A Fair Test



5.3 m

4.1



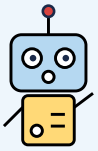
Change one thing. Keep the rest. Write it down. Repeat it.

Puzzles, projects and Multix the guide robot

Ages 9 to 11

This book belongs to _____

Hello. I am Multix.

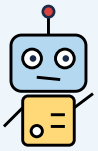


I am a guide robot. I read every instruction very carefully, which is why I sometimes get things wrong. When I make a mistake, see whether you can catch it. I never give away a puzzle answer.

How this book works

Every book in this series asks the same few questions, so once you have read this page you will know your way around.

- **Try it** boxes tell you what to do with a ruler, a cup, some string or a clock. You need nothing from a shop. Everything is at home or at school.
- **What did we keep?** After you simplify something, such as keeping the same conditions in a test or averaging several throws, you say what stayed and what was set aside. Simplifying is fine. Forgetting what you set aside is not.
- **What makes this correction trustworthy?** When you fix or improve an answer, a box says what you checked, what the check gives you reason to trust, and what it does *not* show.
- **Circles** show how a number was made. Page 4 explains how to read them, and Book 1 introduced them.
- **Puzzles** come at the end of each chapter. Try them before you look at the answers on page 44.
- **Slop check** is a page near the end. Slop is hasty, gaudy work that forgot to take care. You will learn to spot it and to fix it.



Here is a fact about me. I tested two paper planes. One had a clip and one did not, but I also launched them from different spots. I decided clips are magic. Chapter 1 will explain why I should be embarrassed.

In this book

1	Two Planes, Two Stories	5
2	One Change at a Time	10
3	Write It So a Stranger Can Repeat It	16
4	Again and Again	22
5	Ice in Wrappings	28
6	What the Evidence Supports	34
⋆	Slop check	38
⋆	Answers	44
⋆	Words, and notes for grown-ups	47

Try it: Gather your kit

You will use these things across the six chapters. Everything is found at home or at school. The measuring skills come from Book 1.

- scrap paper, 20 sheets or more
- a tape measure or long ruler
- some paper clips
- a pencil and a notebook
- a clock or timer showing seconds
- a freezer for ice cubes
- a saucer or small plate, 4 of them
- aluminium foil, a paper towel, a sock
- a kitchen scale (optional)
- a calculator (optional)

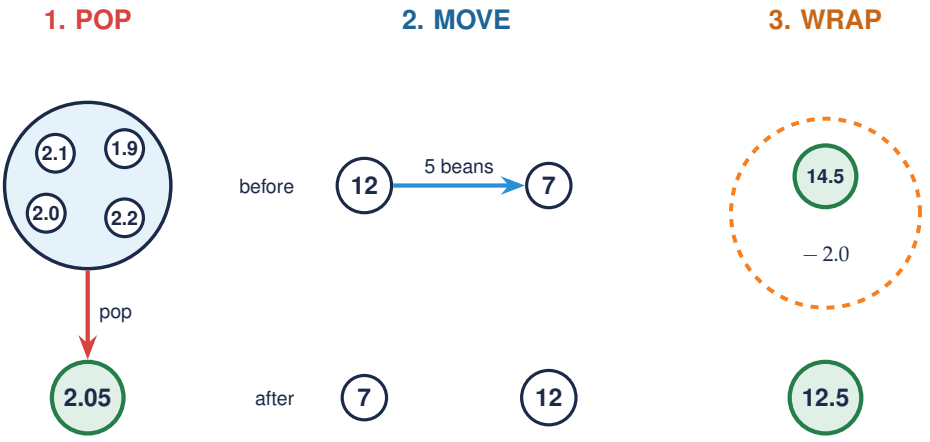
■ **Adult help:** Plane throwing is for open indoor space or outdoors, gently, and never toward faces or pets. Ice cubes melt, so use trays and saucers to keep floors dry. Nothing in this book needs a flame or mains electricity, and no test material should be eaten.

Units in this book

We use the metric system: centimetres (cm), metres (m), grams (g), minutes (min) and seconds (s). There are 10 mm in 1 cm and 100 cm in 1 m. If you are used to inches, one inch is about 2.5 cm. We will use that only in side notes.

Reading the circles

A circle holds a group of things together. There are three things you can do with circles in this series.

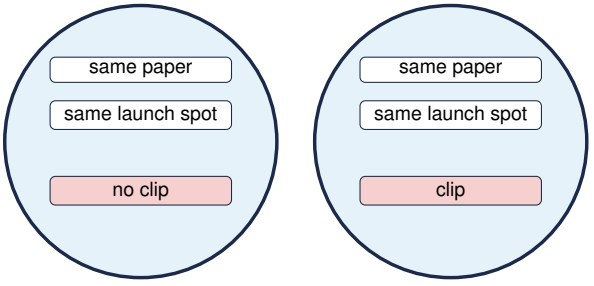


Pop Squash a group down to one value that sums it up, such as an average. A pop is a *summary*, so it always loses something. Every time we pop, we say what the summary keeps and what it loses.

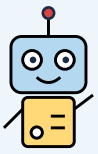
Move An arrow that carries something from one circle to another, such as beans or water. Nothing is summarized. You will see more of moves in later books.

Wrap When you correct something, draw a new dashed circle around the old one and show what changed. The old work stays visible, so anyone can check it.

A test is two circles with the same insides



A fair test is two circles that hold the same things, except for **one** thing you chose to change. If a second thing is different, the result cannot tell the two apart.



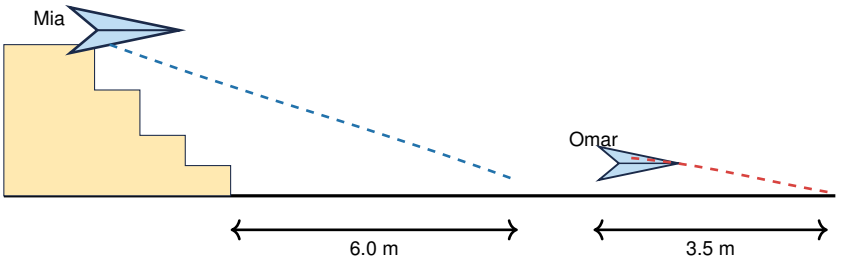
A pop is not a lie, but it is a shortcut. Any time I tell you a pop, I will tell you what it threw away.

1 Two Planes, Two Stories

Why a result is only as good as the test behind it

Mia and Omar each folded a paper plane. Mia is sure that **big wings** fly farther. Omar is not so sure. They decide to test it.

Mia folds a plane from a large sheet and throws it from the top of the stairs. It flies 6.0 m. Omar folds a plane from a small sheet and throws it from the floor of the hall. It flies 3.5 m.



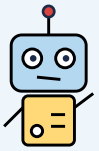
Mia says: “See? Big wings win.” Do you believe her?

Count what was different

Look closely at the two throws. Wing size was one difference. What else was different?

1. The **paper** was a different size, so the plane was also heavier or lighter.
2. The planes were thrown from **different heights**: the stairs and the floor.
3. The planes were **thrown by different people**, and one may throw harder.
4. Each plane was **folded differently**, because two people folded them.

That makes at least five things that changed (counting wing size). If five things changed, the 2.5 m gap could have been caused by any of them. The test cannot say which. We call a test like this an **unfair test**. It is not wrong to run one, but it is wrong to trust it.



I have a plan. I will test whether a clip makes a plane fly farther. I will give a clip to my plane and throw it hard. I will throw the other plane softly, to be kind to it. That sounds like a proper test to me.

Three kinds of things in every test

A test is a question you ask with your hands. To ask it well, sort everything into three groups.

What I change	What I measure	What I keep the same
one thing only	the result I watch	everything else
example: clip or no clip	example: how far it flies	example: paper, spot, thrower

A test is **fair** when you change one thing, measure one result, and keep everything else the same. Scientists call each thing that can change in a test a **variable**, but the three plain names above will do for now.

Try it: Spot the changes

Think of a plant test. Dev wants to find out whether plant food helps beans grow. He gives Plant A plant food and puts it on a sunny windowsill. Plant B gets no plant food and sits in a dark cupboard. After two weeks Plant A is taller.

(a) What did Dev mean to change? _____

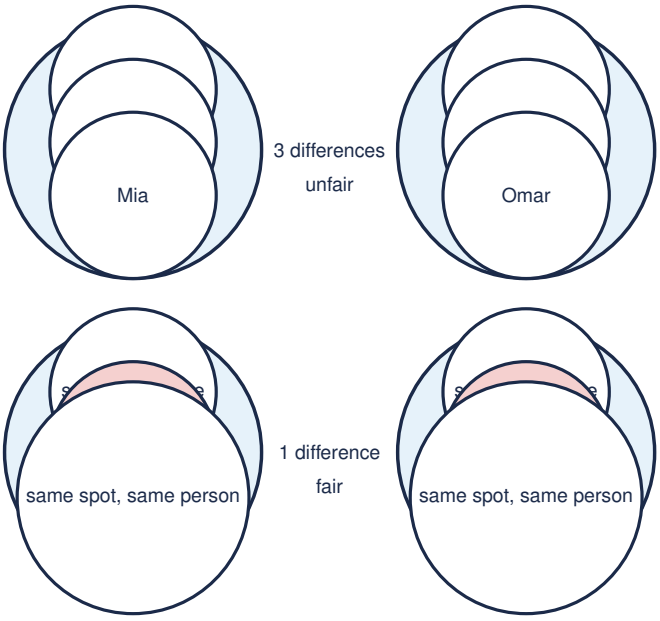
(b) What else is different between the two plants? _____

(c) Is Dev's test fair? Why or why not?

_____

Make the unfair test fair

A repair is easy to describe. Pick the **one** thing you want to learn about. Make it the only difference. Everything else gets copied exactly.



In the fair version both planes use the same size of paper. The wings are different because of how the paper is folded, and that is the one thing being tested. The same person throws both from the same spot at the same height.

What did we keep?

A fair test keeps lots of things the same. **We kept** _____ **the same, and we changed only** _____.

Puzzle 1.1: Count the changes

Three tests are described. For each one, say what is being tested and list any *other* difference.

A "Do heavy balls roll farther? I rolled a tennis ball down a smooth board and a marble down a bumpy board."

B "Does warm water dissolve sugar faster? I put a spoonful of sugar in a cup of warm water and the same spoonful in a cup of cold water, then stirred both for the same time."

C "Does music help plants grow? One plant listened to music in the kitchen. The other had no music and sat in the hall."

Which test is fair? _____ Which has the most hidden differences? _____

Puzzle 1.2: Sort the sentence

Write C (what I change), M (what I measure) or K (what I keep the same) next to each part of this test: "Ana drops a ball from the same height onto sand, grass and a rug. She measures how deep it dents each one with the same ruler."

(a) the surface it lands on _____ (c) the ball and the drop height _____

(b) the depth of the dent _____ (d) the ruler _____

Puzzle 1.3: Who is more careful?

Eli and Nia plan the warm-water sugar test from puzzle 1.1. Eli says: "Both cups must hold the same amount of water, or we will not know whether the temperature or the amount of water made the difference." Nia says: "The cups can hold different amounts of water. We only care about temperature."

Who is more careful? _____

Write one reason, in your own words.

_____

Puzzle 1.4: Fair or unfair?

Mark each test F (fair) or U (unfair). If it is unfair, name one difference.

(a) Two cups of lemon water, one with a spoon of sugar, same lemons, same cups, same stirring. _____

(b) Fast shoes against slow shoes: run one pair early in the morning and the other late at night. _____

(c) Seeds in sand against seeds in soil, same number of seeds, same windowsill, same water. _____

(d) Which battery lasts longer? Brand A runs a small torch. Brand B runs a bigger toy. _____

(e) The same ball dropped from the same height onto sand, grass and a rug, measured with the same ruler. _____

Puzzle 1.5: Multix and the paper towels

Multix wants to know which brand of paper towel soaks up more water. Brand A: a quarter of a sheet in a big bowl. Brand B: a whole sheet in a small cup.

(a) What does Multix mean to change? _____

(b) Name two other differences. _____

(c) How would you fix the test? _____

_____

Puzzle 1.6: Pick the one change

For each question, write the *one* thing you would change.

(a) Do bigger paper planes fly farther? _____

(b) Do ice cubes melt faster in warm water? _____

(c) Does a taller tower of blocks fall over sooner? _____

2

One Change at a Time

Running a real fair test with paper planes

Now it is your turn to run the test properly. The question: **does a paper clip on the nose make a paper plane fly farther?**

First write the three kinds of things down. Writing them is half the work.

I change	a paper clip on the nose: none, or one clip
I measure	how far the plane flies, from the launch line to where it first lands, to the nearest 0.1 m
I keep the same	the paper, the folding, the launch line, the height and the push, the person throwing, the room

Try it: The paper clip test

You need: six sheets of the same scrap paper, one paper clip, a tape measure, a sticky note or tape for the launch line.

1. Fold two planes from two sheets of the same paper. Use exactly the same folds. Hold them side by side to check they match.
2. Put a paper clip on the nose of one plane. Leave the other plain.
3. Stand behind the launch line. Throw each plane **gently, from shoulder height, with the same push**, three times each. Do the throws in a mixed order: plain, clip, plain, clip, plain, clip.
4. After each throw, measure from the launch line to where the plane first touched the floor. Write the distance down at once.
5. Find the average distance for each plane.

	Throw 1 (m)	Throw 2 (m)	Throw 3 (m)	Average (m)
No clip				
Clip				

■ **Adult help:** Throw gently and never toward a person or pet. Pick a clear stretch of floor or a calm day outdoors, and put the clips away afterwards. Small clips are not for mouths.

Why mixed order?

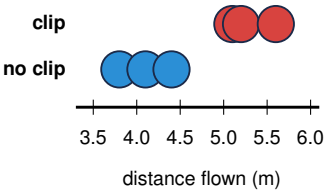
If you throw all the plain planes first and the clipped ones after, your arm may get warmer, tired or better at throwing as you go. That is another difference hiding in the test. Mixing the order spreads that difference across both planes.

Practice numbers

Here are results from a made-up run, so you can see how the numbers work. They are practice numbers, not from a real experiment.

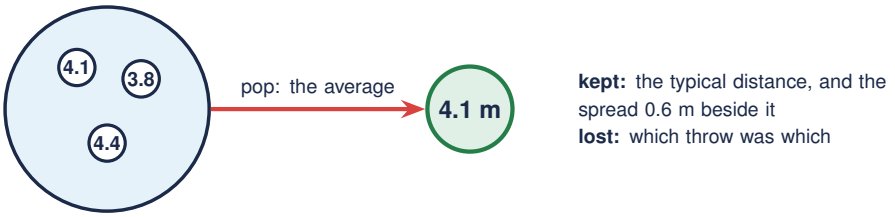
Practice numbers for the paper clip test

No clip: 4.1 m, 3.8 m, 4.4 m. Total $4.1 + 3.8 + 4.4 = 12.3$. Average $12.3 \div 3 = 4.1$ m. Spread $4.4 - 3.8 = 0.6$ m.
Clip: 5.2 m, 5.6 m, 5.1 m. Total $5.2 + 5.6 + 5.1 = 15.9$. Average $15.9 \div 3 = 5.3$ m. Spread $5.6 - 5.1 = 0.5$ m.
Difference of the averages: $5.3 - 4.1 = 1.2$ m.



All three clipped throws went farther than all three plain throws. The two groups do not overlap. That is a clue that the clip really did something.

Three throws become one circle



What makes this correction trustworthy?

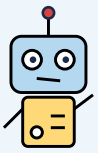
What we checked. We changed one thing, a clip on the nose, kept everything else the same, and threw each plane three times in a mixed order. The groups of throws did not overlap.

What this check gives us reason to trust. If only the clip differed, then a difference in distance is unlikely to come from the paper, the thrower or the spot, so the clip is the likely reason for the gap. Mixing the order also spreads out any tiredness or warming up.

What it does not show. That a clip helps every plane, on every day, or why it helps. It also does not show that three throws are enough. We tested one design of plane and one kind of clip.

What did we keep?

We wrote down what stayed the same. **We kept _____ the same and we set aside _____.**



My own test took 9 seconds. I threw a clip plane hard, 7.5 m, and a plain plane softly, 3.0 m. The clip adds 4.5 m. That is science!

A second round: wing flaps

Now ask a different question with the same method: does **folding the back edge of the wings up** (a small flap) change how far the plane flies? Change nothing but the flaps. Keep the clip off both planes this time, because now the clip is one more thing to keep the same.

Try it: The flap test

Fold two matching planes. Bend the back edge of both wings up by about 1 cm on **one** plane only. Throw each three times in a mixed order from the same line and push. Fill in the table.

	Throw 1 (m)	Throw 2 (m)	Throw 3 (m)	Average (m)
Flat wings				
Flaps up				

What stays the same, and what changes

Use the table to plan every new test. Fill in the blanks for the flap test.

I change	_____
I measure	_____
I keep the same	_____

Puzzle 2.1: Averages

In a clip test, the plain plane flew 3.9 m, 4.2 m and 4.5 m. The clipped plane flew 5.1 m, 5.4 m and 5.7 m.

(a) Plain: total _____ average _____ spread _____

(b) Clip: total _____ average _____ spread _____

(c) Difference of the averages: _____ m

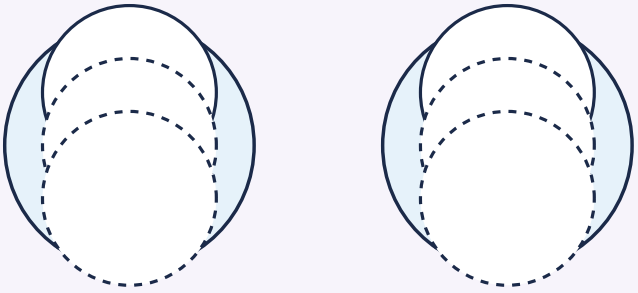
Puzzle 2.2: Plan the test

For each question, say what you would change, what you would measure, and one thing you would keep the same.

	Change	Measure	Keep same
(a) Does a longer ramp make a toy car go farther?	_____	_____	_____
(b) Does warm water melt an ice cube faster?	_____	_____	_____

Puzzle 2.3: Draw the fair circles

You want to test whether the *colour of paper* changes how far a plane flies. Fill in the two circles so the test is fair.



Write in each dashed circle one thing that must be the same.

Puzzle 2.4: Which result should you trust more?

Priya threw each plane once: the clip plane flew 5.0 m and the plain plane flew 4.2 m. Sam threw each plane three times: clip 5.2, 5.6, 5.1 m and plain 4.1, 3.8, 4.4 m.

Whose result should you trust more? _____

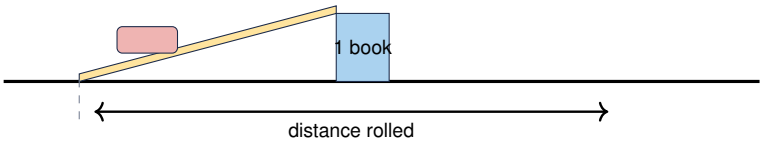
Write one reason.

_____

A third test: a ramp

Try the method on a toy car. The question: **does a higher ramp send a toy car farther along the floor?** Make a ramp from a board or a thick piece of card. Raise one end with one book for the first version and two books for the second.

I change	the height of the top end: one book or two books
I measure	how far the car rolls from the bottom of the ramp to its front wheel when it stops, nearest cm
I keep the same	the board, the car, the floor, where the car starts on the ramp, letting go with no push



Ramp test (practice numbers, not from a real test)

One book: 85, 92, 88 cm. Total $= 265$. Average $265 \div 3 \approx 88$ cm. Spread $92 - 85 = 7$ cm.
Two books: 130, 138, 127 cm. Total $= 395$. Average $395 \div 3 \approx 132$ cm. Spread $138 - 127 = 11$ cm.
Gap: $132 - 88 = 44$ cm, much bigger than either spread.

Try it: The ramp test

Run the test yourself with three rolls of each version in a mixed order.

	Roll 1 (cm)	Roll 2 (cm)	Roll 3 (cm)	Average (cm)
One book				
Two books				

3

Write It So a Stranger Can Repeat It

A test that cannot be repeated cannot be checked

Suppose you ran the paper clip test and found that the clip helped. A friend in another town reads your result. They want to know: is it true? The best answer you can give is not "trust me". It is a card with everything on it, so your friend can run the same test and see for themselves.

That is what scientists do. A result that nobody else can repeat is hard to build on. So the second half of a fair test is **writing down what you kept the same**.

The test card

Here is a card for the paper clip test. Notice that every line tells the reader something they would otherwise have to guess.

Question. Does a paper clip on the nose make a plane fly farther?

I change. A paper clip: none, or one small clip on the nose.

I measure. Distance from the launch line to where the plane first lands, nearest 0.1 m, using a tape measure.

I keep the same.

- Paper: A4 or letter sheets from the same pack.
- Fold: the same dart fold, checked by laying the planes on top of each other.
- Launch: from a marked line, standing still, arm at shoulder height, plane level.
- Push: gentle, as if passing a cup of water. Same thrower for every throw.
- Place: the same hall or yard, no wind, no obstacles.

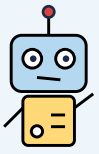
How many times. Three throws each, in the mixed order plain, clip, plain, clip, plain, clip.

What I will report. All six distances, the average of each group, and the spread.

Exact words beat vague words

"Throw it hard" means one thing to you and another to your friend. "Same push" is a start, but "gentle, as if passing a cup of water" is a picture the reader can copy. Whenever a line of the card says something vague, replace it with something someone else could repeat.

Vague	Better
a normal piece of paper	A4 or letter paper from the same pack
throw it from a good spot	from behind a taped line, standing still
throw it the same	arm at shoulder height, gentle, plane level
measure how far it went	nearest 0.1 m from the line to where it first lands
do it a few times	three throws each, in a mixed order



I have written a test card that says “Throw the plane a good way.” I think it is excellent. Anyone can follow it. Can they?

Try it: Write your own card, then trade

Choose a question you can test with things at home. For example: “Does a ramp that is higher make a toy car roll farther?” Fill in the card below. Then give it to a friend or grown-up *without talking*. Can they do your test using only the card?

Question _____

I change _____

I measure _____

I keep the same _____

_____

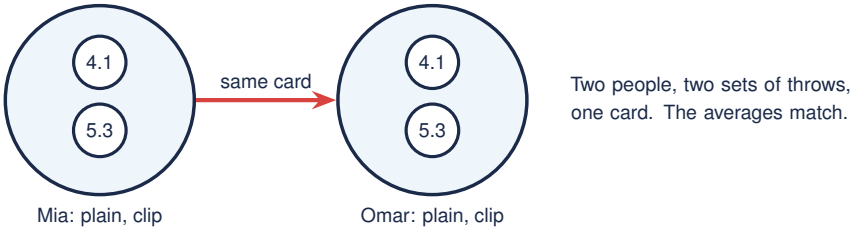
How many times _____

Did the stranger agree?

Omar follows Mia's card exactly. He does not know what she found. Here are his practice numbers, a made-up run.

Omar repeats the test with Mia's card (practice numbers)

No clip: 3.9, 4.3, 4.0 m. Total $3.9 + 4.3 + 4.0 = 12.2$. Average $12.2 \div 3 \approx 4.1$ m.
Clip: 5.0, 5.4, 5.5 m. Total $5.0 + 5.4 + 5.5 = 15.9$. Average $15.9 \div 3 = 5.3$ m.
Mia's averages were 4.1 m and 5.3 m. Omar's are also about 4.1 m and 5.3 m. The result was **repeated**.



What makes this correction trustworthy?

What we checked. Another person followed the card and threw the planes themselves. Their averages, 4.1 m and 5.3 m, matched the first person's.

What this check gives us reason to trust. When a stranger gets about the same answer from the same written conditions, the result is not just one person's luck or habit. Every line of the card was clear enough to follow.

What it does not show. Whether the clip helps other planes, in other rooms, or with other paper. It also does not show *why*. Matching averages show that the test can be repeated, not that the explanation is right.

What did we keep?

A good card lets someone else repeat your test. **We kept** _____ **on the card, and we left off** _____.

Puzzle 3.1: The missing line

Read this test card and find what is missing.

> **Question:** Does the length of a ramp change how far a toy car rolls?
> **I change:** the length of the ramp, 30 cm or 60 cm.
> **I measure:** how far the car rolls after the ramp, nearest cm.
> **I keep the same:** the car, the floor, the person who lets go.

What one important thing is missing from "I keep the same"? _____

Why would it matter? _____

Puzzle 3.2: Put the steps in order

These steps for the paper clip test are in a jumbled order. Write 1 to 6 beside them.

Throw the plain plane, then the clip plane, in a mixed order _____

Fold two matching planes _____

Find the average for each group _____

Write down the distance after each throw _____

Put a clip on one plane _____

Tape a launch line on the floor _____

Puzzle 3.3: Make it exact

Rewrite each vague line so that a stranger could follow it.

(a) "Use a big cup." _____

(b) "Wait a while." _____

(c) "Stir it well." _____

Puzzle 3.4: Two people, two answers

Two classes followed the same card for the clip test. Class A found that the clip plane flew 1.2 m farther. Class B found it flew only 0.1 m farther.
Write two things that could have been *missing* from the card, so the classes did slightly different tests.

_____

_____

Puzzle 3.5: Turn a story into a card

Maya wrote: "I rolled my car down a ramp, then I stacked more books under the ramp and rolled it again. It went farther." Fill in the card so a stranger could repeat her test.

Question _____

I change _____

I measure _____

I keep the same _____

How many times _____

Puzzle 3.6: Same card, different people

Three people followed the same paper clip card. Their averages for the plain plane were 4.1, 4.0 and 6.9 m.

(a) Which person's result is different from the others? _____

(b) Write two questions you would ask that person about how they did the test.

_____

_____

Try it: Swap cards and compare

Give your card from the earlier activity to a friend or grown-up. Ask them to run your test. Write your two averages and theirs below. Do the results agree?

	Version 1 average	Version 2 average
My results		
Their results		

What did the card leave out? _____

A second card: the ice test

Here is the card for the ice test from Chapter 5. Read it, then check that every line could be followed by a stranger.

Question. Which wrapping keeps an ice cube from melting longest?

I change. The wrapping: none, one layer of foil, one paper towel, or one wool sock.

I measure. Minutes until the cube has fully melted, checked every 5 minutes with a timer.

I keep the same.

- Ice: four cubes from the same tray, taken out together.
- Saucer: four identical saucers, side by side on one table.
- Place: away from sunlight and drafts, in the same room.
- Start: all four wrapped before the timer starts.

How many times. Two runs, with new ice each time.

What I will report. Both times, the average for each wrapping, and the spread.

Puzzle 3.7: Two things the card still leaves out

This card says nothing about how *thick* each wrapping is, or how many layers of the sock or towel to use. Write one more line for "I keep the same" that would fix it.

_____

Write one more thing a stranger might wonder about. _____

4 Again and Again

How many throws before you can tell two things apart?

Fair is not the same as finished. Even a fair test can fool you if you throw each plane only once. Every throw is a little different. A gust of air, a different wrist angle or a bad landing can add or take away half a metre.

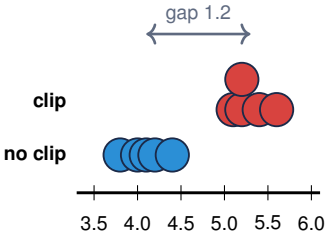
In Book 1 you learned that repeating a measurement and taking the average makes the answer steadier. A fair test needs the same idea. Throw each plane several times. Then compare the **averages**, and compare the **spread**.

Five throws each

Earlier you saw the first three throws of the clip test. Here are all five throws from the same practice run.

Paper clip test, five throws each (practice numbers)

No clip: 4.1, 3.8, 4.4, 4.0, 4.2 m. Total = 20.5. Average $20.5 \div 5 = 4.1$ m. Spread $4.4 - 3.8 = 0.6$ m.
Clip: 5.2, 5.6, 5.1, 5.4, 5.2 m. Total = 26.5. Average $26.5 \div 5 = 5.3$ m. Spread $5.6 - 5.1 = 0.5$ m.
Gap between the averages: $5.3 - 4.1 = 1.2$ m, which is *bigger* than either spread (0.6 and 0.5).



A rule of thumb

When two groups of repeated results are compared, ask: is the gap between their averages bigger than the spread in each group?

- If the gap is **much bigger** than both spreads, the difference is probably real.
- If the gap is **smaller** than the spread, you cannot tell yet. Throw more.

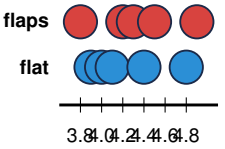
This is only a rule of thumb. It is not a proof, and real scientists use more careful methods. But it stops you from calling a lucky throw a discovery.

When you cannot tell

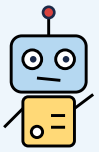
Now compare the flap test. Practice numbers again, five throws each:

Flap test, five throws each (practice numbers)

Flat wings: 4.0, 4.8, 3.9, 4.4, 4.1 m. Total = 21.2. Average $21.2 \div 5 = 4.24$, about 4.2 m. Spread $4.8 - 3.9 = 0.9$ m.
Flaps up: 4.5, 3.8, 4.9, 4.2, 4.3 m. Total = 21.7. Average $21.7 \div 5 = 4.34$, about 4.3 m. Spread $4.9 - 3.8 = 1.1$ m.
Gap between the averages: $4.3 - 4.2 = 0.1$ m. The spreads are 0.9 and 1.1.



The dots for the two groups lie all over each other. The gap, 0.1 m, is much smaller than the spread. The test does not show that flaps change the distance. It does not show that they *do not* change it either. It only says that these ten throws are too few and too wobbly to tell. A fair test can end in "we cannot tell yet", and that is an honest result.



The flaps averaged 4.3 and the flat wings 4.2. So flaps win! I will announce it at lunch.

What makes this correction trustworthy?

What we checked. We threw each plane five times in a mixed order, found each average and spread, and compared the gap between the averages with the spreads.

What this check gives us reason to trust. Comparing a gap to the spread stops one lucky or unlucky throw from deciding the answer. A gap much bigger than both spreads is hard to blame on chance alone.

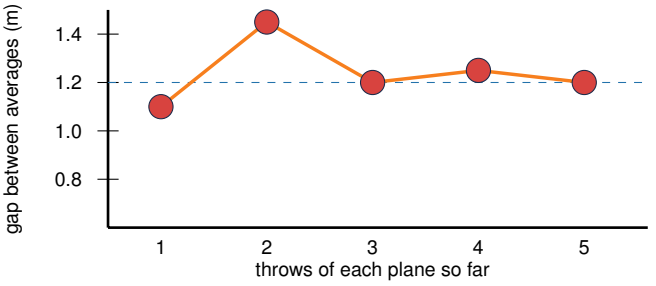
What it does not show. That a bigger gap is always real, or that a gap smaller than the spread means there is no effect. A tiny effect needs many more throws to find. The rule of thumb is a guide, not a proof.

What did we keep?

We pop five throws into one average and a spread. **We kept** _____ **and we set aside** _____.

More throws, steadier answers

In the clip test, how does the average change as you add throws? Watch the running average of the *gap*. It starts rough and settles down.



After one throw each, the gap was $5.2 - 4.1 = 1.1$ m. After two throws each, it was $5.4 - 3.95 = 1.45$ m. After three, it was 1.2 m. After four, it was $5.325 - 4.075 = 1.25$ m, and after five it was back to 1.2 m. The wiggles get smaller as throws are added, so one strange throw matters less and less.

Try it: Throw until it settles

Repeat the paper clip test, or the flap test, for **five** throws each. After every pair of throws, write the running gap in the table. Does the gap settle down?

Throws so far	1	2	3	4	5
Average, plain or flat					
Average, clip or flaps					
Gap					

The lucky throw

What if one throw is far from the rest? Suppose the plain plane flew 4.0, 4.2, 4.1, 4.3 and, once, 7.2 m, because a gust of wind caught it. The average of all five is $23.8 \div 5 \approx 4.8$ m, but four of five throws were close to 4.2. Do not delete the 7.2. Write it down, say why you think it happened, and then also report the average of the other four (4.15, about 4.2). Both numbers tell the reader something.

Puzzle 4.1: Means and spreads

Two groups of five throws are listed. Find each total, average and spread, and the gap.

	Throws (m)	Total	Average
Group X	3.0, 3.3, 3.1, 3.4, 3.2	_____	_____
Group Y	3.9, 4.1, 4.0, 4.2, 3.8	_____	_____

Spread of X: _____ Spread of Y: _____ Gap: _____
Is the gap bigger than both spreads? _____

Puzzle 4.2: Can you tell?

Class A: 5.0, 5.9, 4.4, 5.6, 5.1 m (average 5.2, spread 1.5). Class B: 5.3, 4.5, 5.9, 5.0, 5.4 m (average 5.2, spread 1.4).
The averages are the same. Does this prove the two planes fly equally far? _____
Write one sentence about what you would do next.
_____

Puzzle 4.3: How many more?

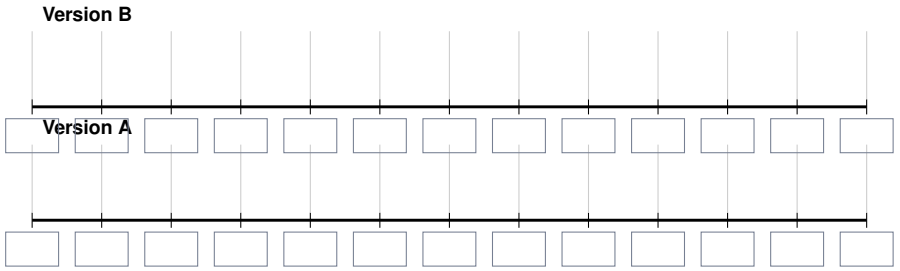
A test gave: plain 4.0, 4.5, 4.3 and clip 4.4, 4.0, 4.7 (metres).
(a) Averages: plain _____ clip _____
(b) Spreads: plain _____ clip _____
(c) What should the next step be: celebrate, give up or throw more? _____

Puzzle 4.4: Hidden difference

A student threw the plain plane five times first, then the clip plane five times. The clip plane flew farther. Name a second thing that could have changed between the two groups, and how a mixed order would help.
_____
_____

Plot your own throws

Put the results of your own clip test or ramp test on the axes below. Use blue for one version and red for the other, with one dot per throw. Write the numbers under the marks that suit your results.



Average of A: _____ Spread of A: _____

Average of B: _____ Spread of B: _____

Gap: _____ Is the gap bigger than both spreads? _____

Puzzle 4.5: Read the dot plot

Dots show how far two paper cups rolled, in cm. Cup A: 40, 42, 41, 43, 44. Cup B: 52, 55, 53, 51, 54.
(a) Average of A _____ average of B _____
(b) Spread of A _____ spread of B _____
(c) Gap _____. Is it bigger than both spreads? _____

My test diary

Use this page for any test you run for fun. Write what you changed, what you measured, and how it turned out.

Day	Question	What I changed	My averages	Could I tell?

5 Ice in Wrappings

A new question, tested with the same method

Everything so far was about paper planes. But the method is not about planes. It works for any question where you can change one thing and measure a result.

Here is a new question. A cold drink at a picnic stays cold if you wrap it. Which wrapping keeps ice from melting longest: **nothing, aluminium foil, a paper towel or a wool sock**?

Plan before you start

Write the three kinds of things before touching any ice. Then think about what could secretly differ.

I change	the wrapping: none, foil, paper towel or wool sock
I measure	how many minutes until the ice cube has melted into water, checked every 5 minutes
I keep the same	ice cube size and shape, starting temperature, saucer, spot in the room, start time, how many layers

Try it: Ice in four wrappings

You need: four ice cubes from one tray, four saucers, foil, a paper towel, a wool sock, a timer, a notebook.

1. Put the four ice cubes on four saucers. Wrap three of them: one in foil, one in a paper towel, one in a sock. Leave one bare.
2. Put all four saucers on a table, away from the sun, side by side. Start the timer when you start the last wrap.
3. Every 5 minutes, peek. Write down when each ice cube has **fully** melted into water.
4. Repeat the whole test a second time with new ice cubes. Compare the two runs.

Wrapping	Run 1 (min)	Run 2 (min)	Average (min)
Bare			
Foil			
Paper towel			
Wool sock			

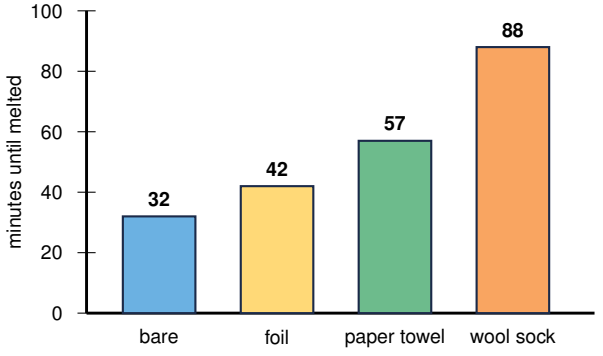
■ **Adult help:** Ice cubes are cold and wet. Let an adult handle the freezer and clean up spills. Do not taste the water or ice.

Practice numbers

Ice in four wrappings (practice numbers, not from a real test)

Wrapping	Run 1	Run 2	Total	Average (min)
Bare	30	34	64	32
Foil	40	44	84	42
Paper towel	55	59	114	57
Wool sock	90	86	176	88

Spread in each pair is 4 minutes. The gaps between neighbours are 10, 15 and 31 minutes.

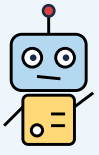


What makes this correction trustworthy?

What we checked. We changed only the wrapping, kept the ice, the saucers, the spot and the start time the same, and ran the whole test twice. The two runs agreed to within 4 minutes for each wrapping.

What this check gives us reason to trust. Repeating the test and getting the same order of wrappings makes a fluke unlikely. The gaps between wrappings (10, 15 and 31 minutes) are much bigger than the 4-minute difference between runs.

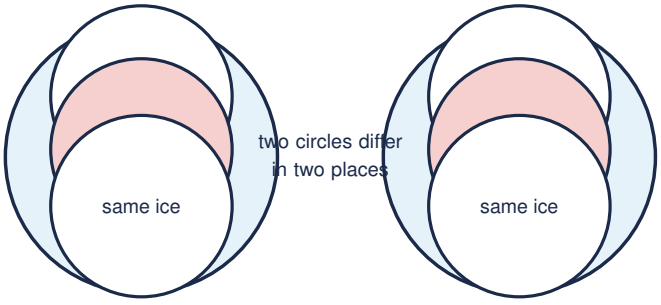
What it does not show. Why the sock wins, or whether it would win with warmer air, bigger ice or a thinner sock. We tested one sock, one foil and one paper towel.



The sock lasted 88 minutes and the foil 42. So wool is twice as good as metal at keeping ice cold. Is that a fair thing to say?

A second change hiding in the test

Look at the wrappings. The wool sock is much **thicker** than the foil. Foil is a thin sheet. So there are really two differences between the sock and the foil: the *material* and the *thickness*. Our test changed both at once. We cannot say whether wool is better than metal, only that this thick sock beat this thin foil.



To find out about the material alone, wrap the foil in several layers until it is as thick as the sock, or compare materials that are the same thickness. That would be a new, better test.

What did we keep?

Our ice test changed the wrapping. **We kept _____ the same, but we also let _____ change by accident.**

Puzzle 5.1: Ice averages

Ice in a foil wrap melted in 44 and 48 min across two runs. Ice in a paper towel melted in 60 and 62 min.

(a) Foil average: _____ min Paper towel average: _____ min

(b) Gap between the averages: _____ min

(c) Spreads: foil _____ paper towel _____

(d) Is the gap bigger than both spreads? _____

Puzzle 5.2: Sponge test

Two sponges were soaked in water and weighed. Sponge A is large and held 52 g of water. Sponge B is small and held 38 g of water. A student says: "Sponge A is a better sponge."

(a) What difference is hiding in this test? _____

(b) How could you make the test fair? _____

(c) A dry sponge weighs 10 g and wet weighs 62 g. How much water did it hold? _____ g

Puzzle 5.3: Spot the second change

Which of these ice tests has a hidden second change? Tick every one that does.

A Four ice cubes from one tray, four wrappings, same table, same start.

B A big ice cube bare and a small ice cube in foil, same table, same start.

C One ice cube in the sun in foil, another in the shade bare.

Tests with a hidden change: _____

Puzzle 5.4: Plan your own ice test

Plan a fair test to find out whether **salted ice** melts faster than plain ice. (Do not taste it.) Fill in the card.

I change _____

I measure _____

I keep the same _____

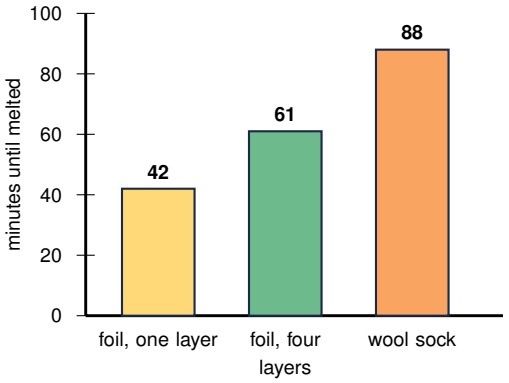
_____

Fixing the hidden change

Earlier we saw that the sock is thicker than the foil. Here is a better test: wrap the foil in **four layers** so it is about as thick as the sock. If thickness is what mattered, the foil should now do much better. Practice numbers, not from a real test:

Thickness test (practice numbers)

Foil, one layer: 42 min. **Foil, four layers:** 61 min. **Wool sock:** 88 min.
Thickness helped: four layers lasted 19 min longer than one layer. But the sock still lasted 27 min longer than the four-layer foil. So in this made-up result, thickness mattered, and the material may matter too.



Try it: Which sponge holds the most?

Cut two different sponges into pieces of the *same size*, say 5 cm by 5 cm. Weigh each dry. Soak each for 10 seconds, let them drip for 10 seconds, weigh again. The water held is wet mass minus dry mass. (No scale? Squeeze each into a cup and compare the water level.)

	Dry (g)	Wet (g)	Water held (g)
Sponge A			
Sponge B			

Sponge test, fair version (practice numbers)

Sponge A: dry 8 g, wet 46 g. Water held $46 - 8 = 38$ g.
Sponge B: dry 10 g, wet 41 g. Water held $41 - 10 = 31$ g.
Same size, same soak, same drip time. A held 7 g more.

Write your own trust box

Pick any test from this book that you have run: the paper clip test, the ramp test, the ice test or your own. Fill in the three lines about it, in your own words.

What makes this correction trustworthy?

What we checked.

_____

_____

What this check gives us reason to trust.

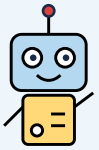
_____

_____

What it does not show.

_____

_____



A good "what it does not show" line is hard to write, and it is the most useful one. Try finishing this sentence: "This test does not show whether…"

Puzzle 5.5: Which trust line is honest?

Three students finished the same sentence for the ice test: "The test does not show…"

A …anything, because ice is cold.

B …whether wool beats foil, because the sock was thicker.

C …that the test was perfect.

Which one is honest and specific? _____

6 What the Evidence Supports

Saying what the test showed, no more and no less

A fair test gives you evidence. Evidence is not the same as a conclusion. This chapter is about the gap between them.

Here are four sentences someone might write after the paper clip test. They are not equally safe.

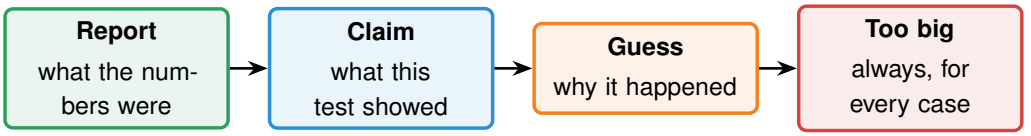
A. "In our test, the clip planes flew 5.3 m on average and the plain planes 4.1 m."

B. "In our test, a clip on the nose made the plane fly about 1.2 m farther."

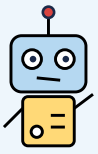
C. "A paper clip makes every paper plane fly farther."

D. "The clip helps because it moves the weight forward."

Sentence A is a plain **report** of the numbers. Sentence B is a careful **claim**: it stays inside the conditions of the test. Sentence C is too big. We only tested one kind of plane, one clip and one thrower. Sentence D gives a *reason*. Our test never looked at weight, so the reason is a guess, even if it is a good guess.



Think of the ladder from Book 1: observed, counted, estimated, concluded. The further right a sentence is, the more you need to *test it separately*. A report needs only the numbers. A guess about why needs a new test.



I tried the clip test on one plane, once, and it flew 8 m. So I can now announce: all clips make all planes fly 8 m. Is that science?

An old, famous fair test

In 1747 a ship's surgeon named James Lind took twelve sailors who all had scurvy and who all ate the same ship's food. He split them into six pairs and gave each pair a different treatment, including cider, vinegar, seawater and oranges and lemons. He reported that the pair who got the oranges and lemons recovered fastest. Lind kept the sailors' food and conditions alike, and changed only the treatment. Twelve people is a very small test by today's standards.

Seven lines for a test report

The five-line report from Book 1 grows by two lines when a test compares things.

1. **Question.** What I wanted to find out.
2. **What I changed and what I measured.**
3. **What I kept the same.**
4. **My readings.** All of them.
5. **My answer.** Averages, spread and the gap.
6. **What I claim, and what I do not.**
7. **What I kept and what it hides.**

A sample report (practice numbers)

1. Question. Does a paper clip on the nose make a paper plane fly farther?
2. Change and measure. I changed the clip (none, or one small clip). I measured the distance from the launch line to the first landing, nearest 0.1 m.
3. Kept the same. The same paper, the same fold, the same thrower, the same launch line and push, the same hall, in a mixed order.
4. Readings (m). No clip: 4.1, 3.8, 4.4, 4.0, 4.2. Clip: 5.2, 5.6, 5.1, 5.4, 5.2.
5. Answer. No clip about 4.1 m (spread 0.6). Clip about 5.3 m (spread 0.5). Gap 1.2 m.
6. Claim. In this test, the clip planes flew about 1.2 m farther. I do not claim this is true for other planes, other clips or other throwers. I do not know why it happened.
7. Kept and hidden. I kept the averages, spreads and conditions. I set aside which throw was which, and the exact flight paths. A different thrower or paper may give different numbers.

Which sentence does the evidence support?

Puzzle 6.1: Safe sentences

In the ice test, wrapping type, average melting times were: bare 32 min, foil 42, paper towel 57, wool sock 88. Mark each sentence S (supported by the test), G (a guess about why, not tested) or B (too big for the test).

(a) "In this test, the ice in the sock lasted longest." _____

(b) "Wool is always the best insulator." _____

(c) "The sock won because it traps air." _____

(d) "The bare cube melted in about 32 minutes on average." _____

Puzzle 6.2: Repair the claim

A student writes: "Flaps make paper planes fly farther." The flap test gave flat 4.2 m and flaps 4.3 m, with spreads of 0.9 and 1.1 m.
Write a better, more honest sentence.

_____

_____

Puzzle 6.3: Find what changed

A test says: "We found that plants given plant food grew 3 cm taller. Both plants were beans and both had the same amount of water. One sat in the sun and the other in the shade." Is this fair? _____ What is the second change? _____

Mixed review

Each puzzle uses something from an earlier chapter.

Puzzle 6.4: Test detective

Read each story. Find the problem with the test.

(a) "We wanted to see whether blue paper planes fly farther than red ones. The blue one flew 5 m and the red one 4 m." (One throw each.) _____

(b) "We tested two kinds of tape for strength. We hung a heavy bag from one and a light bag from the other." _____

(c) "We threw the plain plane ten times. Then we threw the clip plane ten times. We were tired by the end." _____

Puzzle 6.5: Averages again

Plain plane: 3.8, 4.0, 4.2, 4.0, 4.0 m. Clip plane: 5.1, 5.3, 5.0, 5.4, 5.2 m.

(a) Plain: total _____ average _____ (b) Clip: total _____ average _____

(c) Gap: _____ (d) Spreads: plain _____ clip _____

Puzzle 6.6: Which claim?

Two classes ran the ramp test. Class A wrote "A higher ramp sends cars farther." Class B wrote "In our test, a ramp with two books sent the car about 44 cm farther than one book."

Which claim is safer? _____

Why?

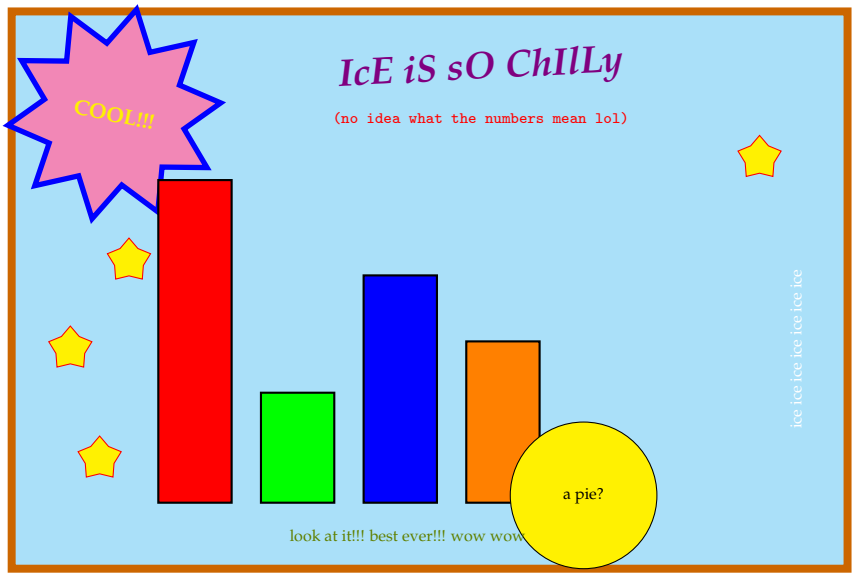
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Slop check

Slop is hasty, gaudy work that forgot to take care. It can be made by a person or a machine, quickly or slowly. What matters is whether the work shows care for three things:

- **Salient:** Is the main point easy to see?
- **Simple:** Is there anything that does nothing but get in the way?
- **Relevant:** Does everything on the page help answer the question?

Here is a hasty poster about the ice test. Study it, then decide what is wrong.



Slop check

(a) What is the poster's main point? Can you tell?

_____

(b) Name two things that are not simple, because they only distract.

_____

(c) Name two things that are missing, so that the bars cannot be read.

_____

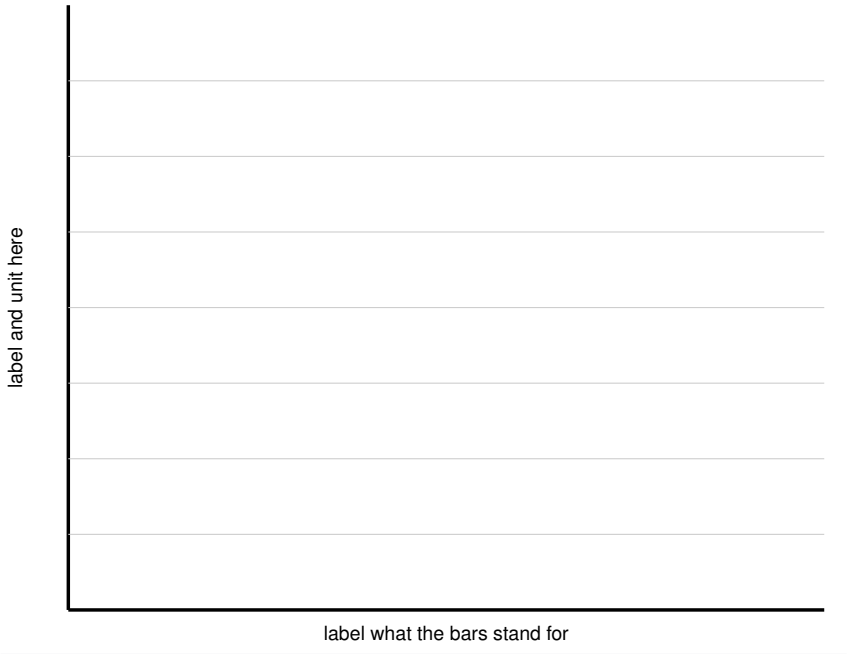
(d) Which part is not relevant to the ice test at all?

_____

Now you try

On a separate sheet, redraw the poster with the same four bars: 32, 42, 57 and 88 minutes. Give it a title that says what was tested, label the bars with the wrapping, label the axis with the unit, pick one or two calm colours, and delete everything that does not help. Check it against the three words: salient, simple, relevant.

Or use the space below for the redraw.



Title:

Design your own fair test

Pick one of these questions, or make up your own: Does a longer ramp send a toy car farther? Does warm water dissolve sugar faster? Does a taller paper tower fall over sooner in a breeze? Fill in the card. Run the test. Then fill in the results table.

Question.

_____

I change.

_____

I measure, with what, to what accuracy.

_____

I keep the same.

_____

_____

How many times, and in what order.

_____

	1	2	3	4	5	Average
Version A						
Version B						

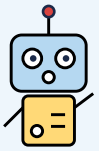
My claim, staying inside the test: _____

_____

What did we keep, across the whole book?

For each chapter, say one thing that was simplified, what was kept, and what was set aside. A sample is done for you.

Chapter	What we simplified	What we kept	What we set aside
1 Two planes	two planes thrown in different ways	the question and the one change we wanted to test	the extra differences that hid the answer
2 One change			
3 Write it down			
4 Again and again			
5 Ice in wrappings			
6 Evidence			



Every test you have run in this book was a simplification of something. A fair test keeps most things the same on purpose, so that one difference can show. Next time a headline says something "works", you will want to ask: compared with what, and what else was different?

Tests to try at home

Here are ten questions you could test with things around the house. Pick three. For each one, write what you would change, what you would measure and one thing you would keep the same.

1. Does a ball bounce higher on a hard floor or on a rug?
2. Does warm water or cold water dissolve a spoon of sugar faster?
3. Does a paper cup or a plastic cup keep a hand-warm drink warm longer?
4. Does a longer string make a pendulum swing more slowly?
5. Does a bigger magnet pick up more paper clips?
6. Does a paper towel or a cloth soak up more water?
7. Does a heavier toy car roll farther down the same ramp?
8. Does a plant by a window grow taller than one in a dim corner? (This takes weeks.)
9. Does stirring dissolve salt faster?
10. Do wide or narrow paper strips hold more coins before they sag?

Number	I change	I measure	I keep the same

Questions I still have

Write a question about testing that this book made you wonder about. Then write how you might find out.

Answers

Try every puzzle first. Where an answer shows a calculation, check the steps, not just the result.

Chapter 1

Multix on the clip test. Throwing one plane hard and the other softly is a second difference, so the test cannot say whether the clip or the push made the gap. **Spot the changes (Dev).** He meant to change plant food. Light also differs: sun against a dark cupboard. The test is not fair. **Keep box.** Kept: the paper, the spot, the thrower, the fold. Changed: only the one thing being tested.

1.1 B is fair. A changes the surface and the kind of ball as well as the weight. C changes the room as well as the music. A has the most hidden differences.
1.2 (a) C (b) M (c) K (d) K.
1.3 Eli. If the amounts of water differ, that is a second change, and then the result cannot say which change mattered.
1.4 (a) F. (b) U: time of day and tiredness differ. (c) F. (d) U: the torch and the toy differ. (e) F.
1.5 (a) The brand of towel. (b) The amount of towel and the size of the container. (c) Use the same amount of each towel in the same container.
1.6 (a) paper size. (b) water temperature. (c) the height of the tower.

Chapter 2

Multix on his own test. He threw one plane hard and the other softly, so the push differs as well as the clip. The 4.5 m gap cannot be blamed on the clip. **Keep box.** Kept: the typical distance and the conditions that stayed the same. Set aside: which throw was which.

2.1 (a) $3.9 + 4.2 + 4.5 = 12.6$, average 4.2 m, spread $4.5 - 3.9 = 0.6$ m. (b) $5.1 + 5.4 + 5.7 = 16.2$, average 5.4 m, spread 0.6 m. (c) $5.4 - 4.2 = 1.2$ m.
2.2 (a) Change the ramp length; measure how far the car rolls; keep the same car, floor and ramp tilt. (b) Change the water temperature; measure the minutes the ice takes to melt; keep the same ice cube, amount of water and starting time.
2.3 Any fair items, such as the same paper size, the same fold, the same thrower, the same launch line.
2.4 Sam. Three throws each is steadier than one. One throw can be a fluke.

Chapter 3

Multix on his card. No. "A good way" is vague, so two people will do different tests. **Keep box.** Kept: the question, the one change, what to measure, what to keep the same and how many times. Left off: opinions and stories about the throws.

3.1 Examples: how the car is released (let go or pushed), the ramp's steepness, or the ramp's surface. A hidden difference there changes the result.
3.2 Throw the plain plane, then the clip plane: 4. Fold two planes: 2. Find the averages: 6. Write down the distance: 5. Put a clip on one plane: 3. Tape a launch line: 1.
3.3 Examples: (a) "Use a cup that holds 250 mL." (b) "Wait 10 minutes, timed with a clock." (c) "Stir 20 times, one turn a second."

3.4 Examples: a different push, a different launch height, a different kind of paper, or measuring to the first landing in one class and to where the plane stopped in the other.
3.5 Question: does a higher ramp send the car farther? Change: the number of books under the ramp. Measure: how far the car rolls past the ramp. Keep the same: the car, the ramp, the floor and letting go without a push. Times: three rolls each, in a mixed order.
3.6 (a) The person who got 6.9. (b) Examples: How hard did you throw? From where? Was the same paper used? How did you measure?
Second card. For example: "Use one layer of each wrapping, about the same thickness, or record the thickness."

Chapter 4

Multix on flaps. A gap of 0.1 m is far smaller than the spreads of 0.9 and 1.1 m, so the test cannot tell. **Keep box.** Kept: the typical distance and the spread. Set aside: which throw was which.

4.1 X: total 16.0, average 3.2, spread 0.4. Y: total 20.0, average 4.0, spread 0.4. Gap $4.0 - 3.2 = 0.8$ m. Yes, the gap is bigger than both spreads.
4.2 No. The averages match, but each spread is about 1.5 m, so the groups overlap heavily. Throw more before you decide.
4.3 (a) Plain $12.8 \div 3 \approx 4.3$, clip $13.1 \div 3 \approx 4.4$. (b) Plain spread 0.5, clip spread 0.7. (c) Throw more: the gap, 0.1 m, is much smaller than the spreads.
4.4 Examples: the thrower may have warmed up, got tired or got better, or the wind may have changed. A mixed order spreads such changes across both planes.
4.5 (a) A: $210 \div 5 = 42$ cm, B: $265 \div 5 = 53$ cm. (b) Both spreads are 4. (c) Gap 11 cm, which is bigger than both spreads.

Chapter 5

Multix on wool and metal. No. The sock is also much thicker than the foil, so material and thickness are both different. **Keep box.** Kept: the ice, the saucers, the spot and the start time. We also let the thickness change by accident.

5.1 (a) Foil $(44 + 48) \div 2 = 46$ min, paper towel $(60 + 62) \div 2 = 61$ min. (b) Gap 15 min. (c) Foil spread 4, paper towel spread 2. (d) Yes, 15 is bigger than 4 and 2.
5.2 (a) The sponges are different sizes. (b) Use sponges of the same size, soaked for the same time. (c) $62 - 10 = 52$ g.
5.3 B and C. In B the ice sizes differ. In C the sun and shade differ as well as the wrapping.
5.4 Change: salt or no salt, with the same amount of salt each time. Measure: minutes until the ice has melted. Keep the same: ice size, saucer, spot and starting time.
5.5 B. It names a real limit of the test. A is silly and C is vague.

Chapter 6

Multix on one throw. No. One throw of one plane cannot show anything about all clips and all planes. **Keep box.** See the recap table.

6.1 (a) S (b) B (c) G (d) S.
6.2 For example: "In this test, planes with flaps flew about 4.3 m and flat planes about 4.2 m. The spreads were bigger than the gap, so the test cannot tell whether flaps matter."

6.3 Not fair. Sun against shade is a second change, as well as the plant food.

6.4 (a) One throw each, so a fluke can decide. (b) The weights differ, so the tape is not the only change. (c) Tiredness grew during the second group, so the order was not mixed.
6.5 (a) $3.8 + 4.0 + 4.2 + 4.0 + 4.0 = 20.0$, average 4.0. (b) $5.1 + 5.3 + 5.0 + 5.4 + 5.2 = 26.0$, average 5.2. (c) Gap 1.2. (d) Spreads 0.4 and 0.4.
6.6 Class B. It stays inside the test and gives the number. **Slop check.** (a) You cannot tell: there is no title, no labels and no units. (b) Many fonts and colours, stars, a pie shape and rotated filler text. (c) Names for the bars and the unit, minutes. (d) The pie, the stars and the jokes.
Redraw. One title, labelled bars and axis with minutes, one or two calm colours and nothing else.

Whole-book table. Chapter 2: simplified a set of throws to an average and a spread, kept the typical distance and the conditions, set aside which throw was which. Chapter 3: simplified a test to a card, kept everything a stranger needs, set aside the thrower's opinions. Chapter 4: simplified five throws to an average and spread, kept the gap compared with the spread, set aside the order of the throws. Chapter 5: simplified a melting cube to a time, kept the minutes and the wrapping, set aside the exact shape. Chapter 6: simplified a test to a report, kept the claim and its limits, set aside the detail of each step.

Words

average the total of the readings divided by how many there are.

claim a sentence that says what a test showed, within the limits of the test.

evidence the numbers and observations a test gives you.

fair test a test where only one thing is changed and everything else is kept the same.

gap the difference between two averages.

keep the same to hold a condition fixed in every trial.

launch line a marked spot that every throw starts from.

repeat to run the same test again, to see whether the result comes back.

spread the biggest reading minus the smallest.

test card a written list of the question, the change, the measurement and everything kept the same.

unfair test a test where more than one thing differs between the versions being compared.

variable something in a test that can change.

Notes for grown-ups

This book teaches that a result is only as good as the test behind it. Each chapter has one activity at a table or on the floor, one worked example with made-up practice numbers (labelled as such), one trust box, one "what did we keep?" prompt, and a page of puzzles.

Materials and safety. Scrap paper, paper clips, a tape measure, a timer, ice cubes, saucers, aluminium foil, a paper towel and a sock. Throw planes gently, never toward faces or pets. Keep clips away from small children. Let an adult handle the freezer, and do not taste test materials.

Check boxes. The trust boxes say what a check shows and what it does not show. Repeating a test shows that it can be repeated, not that the explanation is right. A gap bigger than the spread is only a rule of thumb, not a proof.

The practice numbers in the worked examples (planes, ice, sponges) were made up so that the arithmetic is easy to check. They are not results of real experiments, and real results will differ. The ice example also hides a second change, the thickness of the wrapping, on purpose.

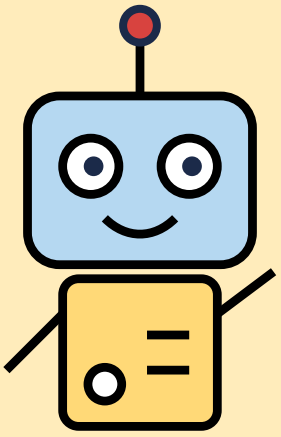
Sources

- R. A. Fisher. *The Design of Experiments*. Oliver and Boyd, 1935. For changing one factor at a time and the logic of fair comparison.
- J. Lind. *A Treatise of the Scurvy*. Edinburgh, 1753. For the 1747 comparison of treatments.
- J. R. Taylor. *An Introduction to Error Analysis*, 2nd edition. University Science Books, 1997. For repeated measurement and spread.

Next in Science

Shadows That Tell Time

A stick in the sun draws a line on the ground, and the line moves. Book 3 shows how light travels, why shadows grow and shrink, and how to build a clock out of a stick, a stone and a sunny day.



Ages 9 to 11 ⋆ Science Book 2